Diagnosing Legal RAG: Bottleneck-Aware Routing of Snap-HyRE

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Abstract

Legal retrieval-augmented generation (RAG) is usually evaluated as a single accuracy number per benchmark, which obscures how it fails. We argue that legal RAG failures are bottleneck-typed: across four legal benchmarks (BarExam, HousingQA, CaseHOLD, LegalBench-SCALR) the useful intervention changes with the dataset, the model, and the slice. We introduce a diagnostic adaptation framework that labels each benchmark from cheap calibration traces (accuracy, gold exposure, conditional accuracy, route disagreement, parse and error counts, and average calls) and routes each row to the cheapest matching intervention among baseline retrieval, legal query rewrite, Snap-HyRE (a two-call HyDE variant in which the model commits to a tentative answer and generates the controlling-rule passage that should support it before retrieval), jurisdiction-scoped metadata filtering, option grounding, conservative verifier policies, disagreement arbitration, and reject/escalate. On Gemma 4 26B over four legal benchmarks, the controller lifts macro accuracy from 71.9% (matched baseline) to 77.9% in calibration at 1.30 average LLM calls per question, and from 71.5% to 77.5% on a held-out slice at 1.54 calls/Q. The lift is uneven and bottleneck-explained: a state-filter plus conservative verifier route adds +14pp on HousingQA and a diverse-HyRE route adds +10pp on CaseHOLD; BarExam exposes route-instability between rewrite and Snap-HyRE, and SCALR's exact-match controller route ties baseline. Fixed Snap-HyRE underperforms baseline on HousingQA (51.5% vs. 60.5%) and CaseHOLD (72.0% vs. 73.0%); we therefore present it as one route inside a portfolio whose value depends on diagnosis, not as a universal recipe. Our contribution is the bottleneck taxonomy, the diagnostic protocol that types each benchmark from its calibration trace, and the routing controller that places generated reasoning on top of the active bottleneck.

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1. Introduction

Legal question answering looks like a natural target for retrieval-augmented generation (Lewis et al., 2020): statutes, holdings, and jurisdictional authority are explicit textual artifacts, and rule exceptions are usually written down somewhere a retriever can fetch. Yet the empirical picture across legal benchmarks contradicts the implicit assumption that better retrieval plus a stronger reader yields uniform gains. Multiple-choice bar questions, yes/no statutory entailment over fifty jurisdictions, holding identification, and reasoning-focused legal retrieval each fail in qualitatively different ways (missing facts, missing jurisdiction, missing option grounding, saturated candidate sets), and a method tuned to one of those failure modes systematically loses on the others. Hallucination audits of deployed legal assistants (Magesh et al., 2024) reinforce that the dominant failure is not a single retrieval gap but a portfolio of distinct, task-shaped breakdowns.

The intellectual move of this paper is to stop asking which retrieval method wins legal RAG and start asking which bottleneck is active on this slice. We instantiate this as a diagnostic adaptation framework: on each benchmark slice we run a small calibration trace, measure a fixed set of signals (accuracy, gold exposure, conditional accuracy, method-pair disagreement, parse and error counts, average LLM calls), label the active bottleneck, and route each row to the cheapest plausible intervention. The portfolio includes baseline retrieval, legal query rewrite, Snap-HyRE (a HyDE-

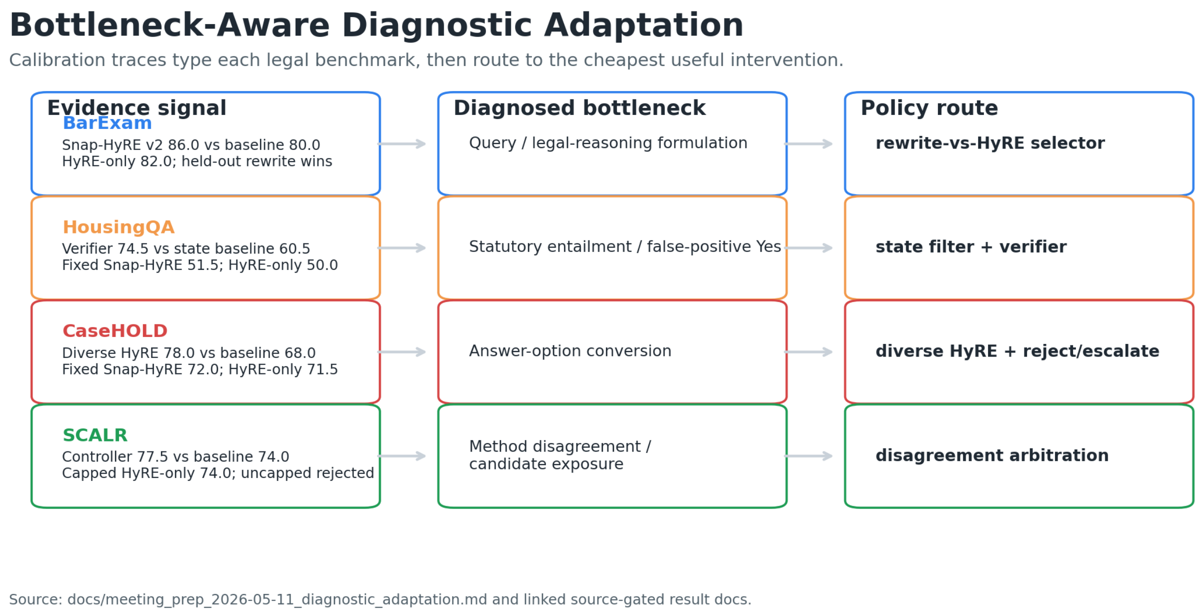

Figure 1. Evidence signal → bottleneck label → routed policy. Each benchmark exposes a different active bottleneck; the controller routes accordingly. BarExam = query/legal-reasoning formulation (route between rewrite and Snap-HyRE); HousingQA = jurisdiction scope + statutory entailment (state filter + verifier); CaseHOLD = answer-option conversion (diverse HyRE); LegalBench-SCALR = method disagreement / candidate exposure (frontier replay).

style (Gao et al., 2023) hypothetical passage conditioned on a snap answer), jurisdiction-scoped metadata filtering, option grounding, a conservative yes/no verifier, disagreement arbitration, and reject/escalate. Unlike prior adaptive-RAG controllers that switch on question complexity or reflection tokens (Asai et al., 2024; Jeong et al., 2024), our bottleneck-aware diagnostic controller routes on evidence-summary signals drawn from a diagnostic decomposition in the spirit of RAGChecker (Ru et al., 2024). Snap-HyRE is one route inside the portfolio, not the controller.

The headline numbers tell a routed story rather than a universal one. On Gemma 4 26B across BarExam, HousingQA, CaseHOLD, and LegalBench-SCALR (Zheng et al., 2025; Guha et al., 2023), the controller lifts macro accuracy from 71.9% at the matched baseline to 77.9% in calibration at 1.30 average LLM calls per question, and from 71.5% to 77.5% on a held-out slice at 1.54 calls per question. The macro lift is bottleneck-explained: a state-filter plus conservative verifier route adds +14pp on HousingQA; a diverse-HyRE route adds +10pp on CaseHOLD; BarExam is route-unstable between legal query rewrite and adaptive Snap-HyRE; SCALR's exact replay route ties baseline while its frontier component reaches the oracle ceiling, exposing a selector-policy gap. The negative controls are part of the evidence: fixed Snap-HyRE underperforms baseline on HousingQA (51.5% vs. 60.5%) and on CaseHOLD (72.0% vs. 73.0%). A universally-applied Snap-HyRE would lose, and the controller captures the gain precisely by withholding it where the active bottleneck is jurisdiction or option conversion rather than query formulation.

We make four contributions. (1) A bottleneck taxonomy for legal RAG with eight named failure modes (retrieval depth, candidate-set saturation, query and legal-reasoning formulation, metadata scope, statutory entailment, answer-option conversion, method disagreement, and parametric floor), each identifiable from cheap calibration signals. (2) A diagnostic adaptation framework that types each slice from those signals and routes each row to the cheapest matching intervention, evaluated on four legal benchmarks. (3) A repositioning of Snap-HyRE as one route, supported by clean negative controls on HousingQA and CaseHOLD and route-instability on BarExam. (4) A held-out cross-model sanity check on Llama 3.3 70B that shows partial transfer: the HousingQA verifier and SCALR frontier lifts hold directionally while BarExam adaptive Snap-HyRE does not.

Section 2 situates the work against RAG, HyDE, adaptive-RAG, and legal-NLP lines. Section 3 formalizes the bottleneck taxonomy and the controller. Section 4 describes datasets, models, and the evaluation protocol. Section 5 reports the calibration and held-out tables, the per-benchmark bottleneck readings, the negative controls, and the cross-model sanity transfer. Section 6 discusses limitations and concludes.

2. Preliminary and Related Work

2.1. Retrieval-augmented generation and HyDE-style query generation

Retrieval-augmented generation conditions a generator on passages retrieved from a non-parametric index, originally to inject up-to-date or domain-specific knowledge into a frozen language model (Lewis et al., 2020; Izacard & Grave, 2021). A practical limitation of the standard recipe is that the literal user query is often a poor embedding target: it is under-specified, missing domain terminology, or phrased differently from the controlling source text. Query-side generation methods address this by turning the query into a richer pseudo-document before retrieval. HyDE synthesizes a hypothetical answer passage and embeds that passage instead of the query (Gao et al., 2023); Query2Doc expands the query with a generated document fragment (Wang et al., 2023); GenRead replaces the retriever with a generator that emits context documents (Yu et al., 2023); and HyQE inverts the direction by generating hypothetical queries for indexed passages (Zhou et al., 2024). Snap-HyRE sits in this family but is conditioned on a snap answer: the model first commits to a tentative legal answer to the question, then writes the controlling rule passage in service of that commitment. The embedding therefore reflects a proposed legal solution rather than a generic pseudo-document, which biases retrieval toward authorities that bear on the specific outcome at issue rather than toward topical neighbors of the surface question.

2.2. Adaptive and agentic RAG

A growing literature treats retrieval as a controllable action rather than a fixed pipeline stage. FLARE retrieves only when the generator's next-token confidence dips (Jiang et al., 2023); Self-RAG trains the generator to emit reflection tokens that gate retrieve, critique, and rewrite steps (Asai et al., 2024); CRAG grades retrieved evidence and applies corrective web search when the retrieved set is judged insufficient (Yan et al., 2024); Adaptive-RAG routes a question to a no-retrieval, single-step, or multi-step pipeline based on a learned question-complexity classifier (Jeong et al., 2024); Adaptive HyDE thresholds the HyDE retriever on per-query confidence to skip generation when retrieval already suffices (Lei et al., 2025); Speculative RAG runs a small drafter against a large verifier to amortize generation cost (Wang et al., 2025); and Iter-RetGen (Shao et al., 2023) together with IRCoT (Trivedi et al., 2023) interleave retrieval with intermediate reasoning steps so that later queries condition on earlier ones. Our bottleneck-aware diagnostic controller differs in two ways. First, we do not route by question complexity or by trained reflection tokens; we route by bottleneck type measured from a calibration trace on each benchmark slice. Second, the intervention space is heterogeneous (rewrite, Snap-HyRE, state-filtered retrieval, conservative verifier, disagreement arbitration, and reject/escalate) rather than retrieve-depth or retrieve-or-not variants of the same skeleton. Source-gated diagnostic measurement, with the bottleneck labeling and intervention selection it implies, is the contribution.

2.3. Legal NLP benchmarks and legal RAG

Legal evaluation has matured around a small set of multi-task and reasoning benchmarks: LegalBench (Guha et al., 2023), which includes the SCALR holdings task we evaluate; CaseHOLD (Zheng et al., 2021), the five-option holding-completion benchmark; and the reasoning-focused legal retrieval benchmark introducing BarExam and HousingQA (Zheng et al., 2025). These are paired with large legal corpora such as LexGLUE (Chalkidis et al., 2022) and Pile of Law (Henderson et al., 2022). Recent work targets retrieval-aware evaluation directly: LegalBench-RAG factors out retrieval from generation on LegalBench prompts (Pipitone & Alami, 2024), and Legal RAG Bench extends the protocol across additional legal tasks (Butler & Butler, 2026). The reliability picture is more sobering: the Stanford hallucination audit of commercial legal AI products documents substantial fabrication and miscitation (Magesh et al., 2024), showing that legal RAG cannot be summarized by a single accuracy number; and Vaddi (2026) reports that adding legal retrieval helps or hurts depending on the task. Our work names that "help-or-hurt" pattern as a bottleneck signature rather than as noise, and operationalizes routing across heterogeneous interventions accordingly.

2.4. Diagnostic evaluation of RAG

Several frameworks now evaluate RAG without per-question gold answers. RAGAS scores faithfulness, answer relevance, and context relevance using LLM judges (Es et al., 2024); ARES trains lightweight evaluators against synthetic judgments and includes prediction-powered confidence intervals (Saad-Falcon et al., 2024); and RAGChecker decomposes errors along retrieval-side and generation-side axes for fine-grained diagnosis (Ru et al., 2024). A related thread documents that even well-retrieved context is underutilized when the relevant passage falls in the middle of a long prompt (Liu et al., 2024). These tools measure RAG failure modes but do not route among interventions; our bottleneck-aware diagnostic controller closes that loop by mapping the same kind of diagnostic signal onto a concrete choice of policy per row.

3. Method: Bottleneck-Aware Diagnostic Adaptation

3.1. Overview

The diagnostic adaptation framework has three moving parts that compose into a single inference-time pipeline. (1) A calibration trace is run once per benchmark slice: a small fixed set of candidate routes is evaluated on rows 0–199 of each dataset, producing per-method accuracy, gold-passage retrieval rate, gold-conditional accuracy, per-row disagreement across routes, parse/error counts, and average LLM-call cost. (2) A bottleneck label is assigned to the slice from a fixed signal set (Section 3.4); each label corresponds to a known failure mode and a known matching intervention. (3) At inference, the bottleneck-aware diagnostic controller performs per-row routing to the cheapest intervention in our portfolio whose preconditions match the slice's bottleneck label and the row's diagnostic signature. The full pipeline is therefore calibrate → label → route, with retrieval, reranking, and embedding shared across all routes. We emphasize at the outset that the bottleneck-aware diagnostic controller is a rule and evidence-summary policy over the calibration trace, not a learned router: there is no policy network, no reflection-token training, and no question-complexity regression. The mapping from signals to intervention is auditable in a small table and is reported in Section 3.5.

3.2. Intervention portfolio: the method ladder

The candidate routes form an inherited method ladder in which each step adds exactly one capability over the step below it, so that a negative result at a higher rung points at a specific intervention rather than at an under-specified bundle. Figure 2 shows the ladder; Table 1 lists the LLM-call cost of each route. The diagnostic bottleneck-aware diagnostic controller chooses among these per row; the ladder is not run end-to-end on any single question.

Every route in Table 1 draws from the same dense encoder, BM25 hybrid pool, cross-encoder reranker, and prompt sanitizer; they differ only in (a) what text is used as the retrieval handle, (b) the metadata predicate, if any, and (c) the final-synthesis prompt. This isolation is what makes the ladder diagnostic: a route winning over its immediate neighbor identifies the single capability that closed the gap.

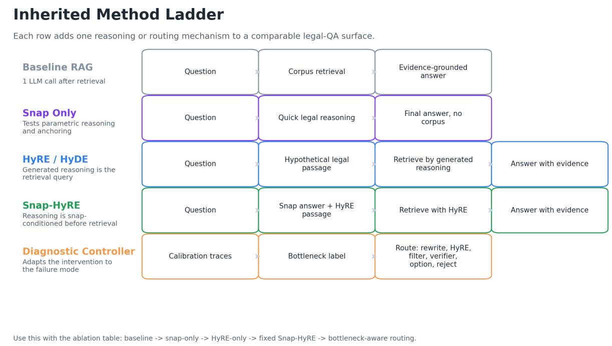

Figure 2. Method ladder. Each step adds one capability so negative results remain interpretable. The diagnostic controller picks per row.

3.3. Snap-HyRE: snap-conditioned hypothetical reasoning

Snap-HyRE is the central new route in the portfolio and the route most sensitive to the active bottleneck. In a single LLM call, the model is prompted to produce two clearly labeled blocks: a ## Reasoning snap answer, in which it commits to a candidate answer based on parametric knowledge alone; and a ## Passage block stating the controlling rule, doctrine, statutory test, fact pattern, or legal principle relevant to resolving the question. The ## Passage block is then embedded with the same dense encoder used to embed the corpus and used as the retrieval handle for dense+BM25 hybrid retrieval and cross-encoder reranking. The final synthesis call sees the retrieved corpus passages plus the original question; the snap answer itself is not handed to final synthesis verbatim, only mediated through the passages that the ## Passage block successfully retrieved.

Two distinctions matter. Against standard HyDE (Gao et al., 2023), the hypothetical document is conditioned only on the question and is asked to look like an answer; against Snap-HyRE, the hypothetical document is conditioned on a snap-committed answer and is asked to state the rule that resolves the question. The architectural justification is brief: an answer-anchored prompt forces the passage toward rule-like, retrieval-shaped text ("The rule against perpetuities voids any interest that may not vest within twenty-one years of a life in being") rather than toward a paraphrase of the question stem, which is the failure mode of pure HyDE on multiple-choice legal items where the stem already carries most of the lexical surface.

Table 1. Method ladder used as the intervention portfolio. Counts refer to LLM generation calls per question; retrieval, embedding, and reranking are not counted.

Route	Calls	Capability added
rag_simple	1	baseline dense+rerank retrieval
rag_state_filter	1	jurisdiction-scoped metadata filter
snap_only_in_final	2	snap reasoning anchors final synthesis
rag_rewrite	2	legal query rewrite before retrieval
rag_hyde	2	HyDE hypothetical passage as retrieval handle
rag_snap_hyde_2call	2	fixed Snap-HyRE (snap-conditioned passage)
adaptive_snap_hyre_v2	2	BarExam route: rewrite/HyRE selector
adaptive_snap_hyre_housing_verifier	2	state filter + yes/no verifier
adaptive_snap_hyre_diverse	2	diverse Snap-HyRE passages for option spread
adaptive_snap_hyre_disagreement_majority_prior	2	disagreement arbitration with majority prior
adaptive_snap_hyre_frontier	2	frontier replay across HyRE-family rows

We do not claim Snap-HyRE is universally better than baseline RAG. The calibration evidence in Section 5 of the paper shows fixed Snap-HyRE underperforming baseline on HousingQA (51.5% vs. 60.5%) and CaseHOLD (72.0% vs. 73.0%); the corresponding HyRE-only controls are weaker still (50.0% on HousingQA, 71.5% on CaseHOLD). Snap-HyRE is therefore positioned as a route inside the portfolio: it lifts BarExam, where the active bottleneck is query/legal-reasoning formulation, but loses on slices where the active bottleneck is jurisdiction scope or answer-option conversion. This is the precise sense in which the controller's gain is bottleneck-explained: Snap-HyRE is selected where it helps and withheld where the negative controls predict it will hurt.

3.4. Diagnostic signals and bottleneck labels

For each candidate route on a calibration slice we record a compact signal vector: route accuracy and the paired delta over the matched baseline; gold-passage retrieval rate and gold-conditional accuracy, where the dataset exposes gold passages; per-row disagreement (the fraction of rows on which two routes give different predicted answers, and the row-level oracle ceiling of picking the best of the two); unparseable-answer and engine-error counts; and average LLM calls per question. The same signals are computed for a top-1 vs. top-5 retrieval-depth ablation on the baseline, since the depth signature is the cheapest discriminator between a depth bottleneck and a parametric floor.

The eight bottleneck labels in Table 2 exhaust the failure modes we observed across the four legal benchmarks. Each row pairs a diagnostic signal with the single matching intervention that closes the gap, and names the benchmark on which that bottleneck dominates. The labels are not mutually exclusive at the row level (a single HousingQA row can be both jurisdiction-scoped and a false-positive yes), but they are mutually exclusive as slice-level labels by construction: the controller picks the dominant signal.

3.5. The bottleneck-aware controller

The bottleneck-aware diagnostic controller is a hand-specified rule policy parameterized by the calibration trace. Algorithm 1 gives the procedure end to end.

Algorithm 1 Bottleneck-aware diagnostic adaptation.

Require: benchmark slice S (rows 0–199); intervention portfolio $\mathcal{R}$ from Table 1; bottleneck taxonomy $\mathcal{B}$ from Table 2 (label $\to$ matching-route subset).

1: Calibration. For each route $r \in \mathcal{R}$, run r on S and record the signal vector $s_S = (\text{acc}_r, \text{gold-hit}_r, \text{cond-acc}_r, \text{pair-disagree}_{r,r'}, \text{parse-err}_r, \text{empty-retr}_r,$

2: Label. For each bottleneck $b \in \mathcal{B}$, compute the signal magnitude $\sigma_b(s_S)$ defined in Appendix B (e.g., $\sigma_{\text{retrieval-depth}} = \text{acc}_{\text{top-5}} - \text{acc}_{\text{top-1}}$, $\sigma_{\text{option-conv}} = \text{gold-hit} - \text{cond-acc}$). Assign $\ell^\star = \arg\max_b \sigma_b(s_S)$.

3: Route selection. Among routes whose preconditions match $\ell^\star$, pick the cheapest by average LLM calls: $r^\star = \arg\min_{r \in \mathcal{R}_{\ell^\star}} \text{calls}_r$.

4: Per-row downgrade. For each held-out row i, if the snap fails to parse, the state filter returns zero candidates, or two cheap routes disagree on the answer, swap $r^\star$ for a slice-specific fallback (typically rag_simple or rag_state_filter).

5: Inference. Apply $r^\star$ (or the fallback) to each row in the held-out slice; report accuracy and average calls.

Three properties matter. (i) The routing rule is a table lookup composed with two arg-ops; there is no policy network, no reflection-token training, and no gradient flowing into the rule. (ii) The slice-level label $\ell^\star$ is fixed once calibration is done; only the per-row downgrade step (step 4) consults the held-out row itself,

Table 2. Bottleneck taxonomy. Each row pairs a diagnostic signal computed from the calibration trace with the matching intervention from the method ladder, and names the benchmark on which that bottleneck dominates.

Bottleneck	Diagnostic signal	Intervention	Dominates on
Retrieval depth	Catastrophic top-1 vs. top-5 drop on baseline	Larger k; smarter passage selection	Depth-sensitive datasets
Candidate-set saturation	Top-1 collapses; top-5/top-10 tie; gold-hit rises but answer flat	Restrict to small reranked candidate set	LegalBench-SCALR
Query / legal-reasoning formulation	Rewrite vs. HyRE asymmetry; snap reasoning lifts more than HyDE retrieval alone	rag_rewrite or adaptive_snap_hyre_v2; route between them	BarExam
Metadata / jurisdiction scope	Empty-retrieval or near-zero state-match rate; explicit state filter beats generic top-k	rag_state_filter	HousingQA
Statutory entailment / false-positive yes	Verifier route lifts over plain state filter; fixed Snap-HyRE underperforms baseline	Conservative yes/no verifier on top of state filter	HousingQA
Answer-option conversion	Gold-retrieval gains do not convert to answer gains (CaseHOLD gold-hit 16.0%→47.0%, EM 69.5%→72.0%, $p = 0.44$)	Diverse Snap-HyRE; direct option-table is a clean negative	CaseHOLD
Method disagreement / candidate exposure	High disagreement across HyRE-family rows; row-level oracle headroom unrealized by simple ensembles	Disagreement arbitration or frontier replay selector	LegalBench-SCALR
Parametric floor / depth-flat	Top-1 $\approx$ top-5 $\approx$ llm_only; snap-only matches RAG	Trust priors; use snap reasoning to anchor, not to retrieve	BarExam

and it only downgrades to the fallback. (iii) Every routing decision is auditable: a reader can read $\sigma_b(s_S)$ off the calibration trace and reconstruct the chosen route. This sits in deliberate contrast with Adaptive-RAG (Jeong et al., 2024), which learns a classifier over question complexity, and Self-RAG (Asai et al., 2024), which trains reflection tokens to gate retrieval. Both prior controllers fix the retrieval pipeline and switch on a property of the question; our bottleneck-aware diagnostic controller also switches on a property of the benchmark's failure mode, which is what enables it to withhold Snap-HyRE on HousingQA and CaseHOLD where fixed Snap-HyRE is a negative control.

Empirically, this controller lifts macro accuracy from 71.9% to 77.9% on the calibration slice at 1.30 average LLM calls per question, and from 71.5% to 77.5% on the held-out slice at 1.54 calls, both at the sub-2-call budget set by the method ladder; the per-benchmark decomposition and the cross-model sanity transfer are reported in Section 5.

4. Experimental Setup

Benchmarks. We evaluate on four legal benchmarks that together exercise distinct retrieval bottlenecks. BarExam (Zheng et al., 2025) provides 1,195 bar-exam multiple-choice questions with fact patterns and aligned legal passages. HousingQA (Zheng et al., 2025) consists of yes/no statutory entailment questions over state housing statutes, where the jurisdiction is recoverable from a state metadata field on the statute corpus. LegalBench-SCALR (Guha et al., 2023) contains 571 Supreme Court holding-selection questions with five candidate holdings. CaseHOLD (Zheng et al., 2021) pairs court-opinion contexts with five candidate holdings. Two notes on scope: (i) MuSiQue is excluded from this submission and the legal-only scope is deliberate; (ii) all runs use deterministic seed 42, with calibration on rows 0–199 ($N = 200$) and a held-out slice on rows 200–249 ($N = 50$) per dataset.

Models. The four-benchmark calibration uses Gemma 4 26B-A4B served through OpenRouter (provider key or-gemma4-26b). Cross-model sanity runs use Groq Llama 3.3 70B (groq-llama70b). Paired comparisons hold the model and the question subset fixed within each benchmark so that every delta is computed over the same rows.

Retrieval stack. We use a hybrid retrieval pipeline that combines BM25 lexical scoring with dense embeddings from Alibaba-NLP/gte-large-en-v1.5 (1024-dimensional, 8,192-token context) over per-corpus Chroma indexes. A cross-encoder reranker (cross-encoder/ms-marco-MiniLM-L-6-v2) selects the final top-k from a candidate pool of size $3k$. BM25 is skipped for collections exceeding one million documents, so HousingQA's statute index is dense-only. Multi-query retrieval pools candidates across all query variants and reranks them against the primary query.

Calibration and held-out protocol. On rows 0–199 of each benchmark we run the inherited method ladder: rag_simple (or rag_state_filter on HousingQA), snap_only_in_final, rag_rewrite, rag_hyde, rag_snap_hyde_2call, and the bottleneck-specific bottleneck-aware diagnostic controller routes. Per-row signals (accuracy, gold-passage retrieval, conditional accuracy, route disagreement, parse and error counts, and average LLM calls) are computed from the detail logs of those runs. On rows 200–249 the controller picks routes from the calibration signals; we report both the selected route per row and individual route accuracies for ablation. Reported metrics are multiple-choice or yes/no accuracy, McNemar p-values for paired comparisons, and average LLM calls per question. Larger-N evaluations ($N \geq 500$) are in progress on all four benchmarks; we report N=200 calibration and N=50 held-out as our primary numbers and note the larger-N expansion as future validation.

Statistical reporting. All paired deltas use exact McNemar tests with reported b/c discordance counts and 95% bootstrap confidence intervals from 2,000 resamples. We do not promote single-row absolute accuracies as primary claims; the headline number is the macro lift across the four legal benchmarks at fixed average-call budget.

5. Analysis

This section reads the calibration and held-out results through the bottleneck lens. We first state the macro headline and the calibration, held-out, and macro-vs-cost views (Figures 3–5 and Tables 3–4). We then walk each benchmark in turn, isolate the negative controls where fixed Snap-HyRE and HyRE-only routes lose to the matched baseline, and close with a cross-model sanity check on Llama 3.3 70B.

5.1. Main result: controller vs. baselines

The bottleneck-aware diagnostic controller lifts macro accuracy from 71.9% to 77.9% (+6.0pp) on the four-benchmark calibration slice at an average of 1.30 LLM calls per question, and from 71.5% to 77.5% (+6.0pp) on the held-out slice at 1.54 calls per question. The held-out lift is concentrated in HousingQA verifier routing (+14pp) and CaseHOLD diverse Snap-HyRE (+10pp); BarExam and SCALR contribute no lift on this slice and instead expose routing-policy headroom that we discuss in the per-benchmark readings below.

Inherited Ablation: Calibration Portfolio

Controller routes improve macro accuracy while using fewer calls than preselected HyRE-family routes.

Model & Method	BarExam	HousingQA	CaseHOLD	SCALR	Avg.	Calls	Gate
Gemma 4 26B + matched baseline	80.0	60.5	73.0	74.0	71.9	1.00	N=200
+ snap-only reasoning	85.5	55.0	72.5	72.5	71.4	2.00	N=200
+ legal query rewrite control	82.0	58.0	72.0	76.0	72.0	2.00	mixed N
+ preselected HyRE-family route	86.0	63.5	73.5	76.0	74.8	2.00	N=200
+ diagnostic controller routes	**86.0**	**74.5**	**73.5**	**77.5**	**77.9**	**1.30**	**N=200 / replay**

Sources: docs/diagnostic_controller_portfolio_comparison_2026-05-10.json and docs/snap_only_controls_2026-05-11.json.

Figure 3. Calibration ablation (Gemma 4 26B, N=200 per benchmark). The diagnostic controller leads the matched-baseline, snap-only, query-rewrite, and preselected-HyRE rows at 1.30 average LLM calls per question.

Method	BarExam	HousingQA	CaseHOLD	SC
Matched baseline	80.0	60.5	73.0	[illegible]
+ snap-only reasoning	85.5	55.0	72.5†	[illegible]
+ legal query rewrite*	82.0	58.0	72.0	[illegible]
+ preselected Snap-HyRE route	86.0	63.5	73.5	[illegible]
+ diagnostic controller	86.0	74.5	73.5	[illegible]

Table 3. Calibration ablation on Gemma 4 26B over BarExam, HousingQA, CaseHOLD, and LegalBench-SCALR; rows 0–199 per benchmark ($N = 200$). Matched baseline is rag_simple on BarExam/CaseHOLD/SCALR and rag_state_filter on HousingQA. The controller dominates every fixed policy at 1.30 average calls per question, below the 2.00 calls spent by the snap-only, rewrite, and preselected-HyRE rows. *Query rewrite is $N = 50$ on HousingQA/CaseHOLD/SCALR and $N = 200$ on BarExam (treated as a control, not a controller route). †CaseHOLD snap-only is reported from a length-capped generation pass.

How to read these tables. Calibration is N=200 per benchmark, so a one-row swing is 0.5pp. Held-out is N=50, so a one-row swing is 2pp, and a +10pp lift corresponds to five rows. We therefore treat held-out absolute deltas as directional rather than distributional: the +14pp HousingQA and +10pp CaseHOLD lifts are larger than reasonable McNemar bands at N=50 but a single seed cannot rule out a ±2pp shift on the BarExam and SCALR ties. The macro lift is robust to per-benchmark perturbations of up to one row because two of the four benchmarks lift cleanly.

The cost dimension is load-bearing for the headline. Snap-only, legal query rewrite, and the preselected-HyRE family each spend 2.00 average LLM calls per question across the four-benchmark calibration slice and achieve 71.4, 72.0, and 74.8 macro accuracy respectively. The controller's 77.9 macro accuracy comes at 1.30 average calls, below all three fixed policies, because it routes the cheapest plausible intervention

Inherited Ablation: Held-Out Slice

Held-out rows validate the routing story, with HousingQA and CaseHOLD carrying the clearest gains.

Model & Method	BarExam	HousingQA	CaseHOLD	SCALR	Avg.	Calls	Gate
Gemma 4 26B + held-out baseline	76.0	62.0	68.0	80.0	71.5	1.00	rows 200-249
+ legal query rewrite	90.0	58.0	76.0	78.0	75.5	2.00	same rows
+ selected diagnostic routes	**76.0**	**76.0**	**78.0**	**80.0**	**77.5**	**1.54**	**same rows**

Sources: docs/heldout_controller_eval_2026-05-10.json and docs/heldout_query_rewrite_2026-05-10.json.

Figure 4. Held-out check (rows 200–249, $N = 50$ per dataset). Matched baselines average 71.5%, query rewrite 75.5%, selected diagnostic routes 77.5%. The held-out lift is concentrated in HousingQA verifier routing (+14pp) and CaseHOLD diverse HyRE (+10pp); BarExam and SCALR tie baseline on the selected route.

Method	BarExam	HousingQA	CaseHOLD	SCALR	Macro	Calls
Held-out baseline	76.0	62.0	68.0	80.0	71.5	1.00
+ legal query rewrite	90.0	58.0	76.0	78.0	75.5	2.00
+ selected diagnostic routes	76.0	76.0	78.0	80.0	77.5	1.54

Table 4. Held-out evaluation on rows 200–249 ($N = 50$ per benchmark). The controller route selected from calibration ties the matched baseline on BarExam and SCALR, lifts HousingQA by +14pp via the verifier route, and lifts CaseHOLD by +10pp via diverse Snap-HyRE. Pure query rewrite is a strong but uneven control: it dominates BarExam (90.0) while losing 4pp on HousingQA.

per row, including single-call replay and state-filter retrieval where the calibration signature does not warrant a HyRE expansion. The controller is therefore a Pareto-improvement over the inherited method ladder rather than a budget trade. Figure 5 makes this concrete: every preselected fixed-policy row sits below the controller on accuracy and to its right on average calls.

5.2. Per-benchmark bottleneck diagnostics

Table 5 summarizes one row per benchmark (active bottleneck, diagnostic signal, chosen route), and the four paragraphs below walk the calibration evidence for each.

BarExam (query / legal-reasoning formulation, route-unstable). The calibration ladder reads rag_simple 80.0, snap_only 85.5, rag_rewrite 82.0, rag_hyde 82.0, rag_snap_hyde_2call 84.5, and the controller route 86.0. The gap between snap reasoning (85.5) and HyDE retrieval alone (82.0) is the diagnostic signature of query/legal-reasoning formulation: the lift comes from the snap answer's framing of the legal rule, not from the HyDE-style passage expansion in isolation. The controller therefore picks adaptive

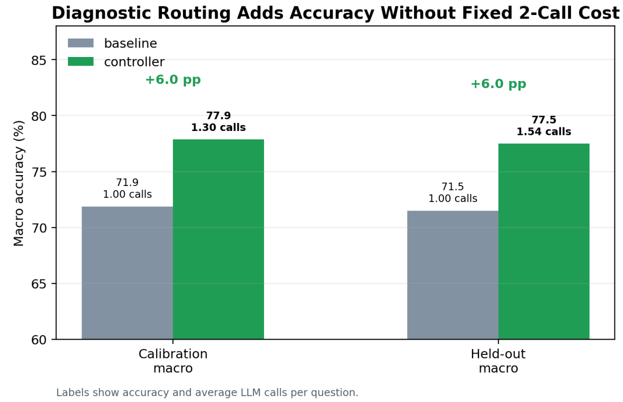

Figure 5. Macro accuracy vs. average LLM calls per question. The controller dominates the matched-baseline and preselected-HyRE points while spending fewer average calls than fixed 2-call methods, because it routes some questions through cheaper replay/verifier policies.

Snap-HyRE v2, which dominates the calibration slice. The held-out slice flips: rag_rewrite reaches 90.0 while the controller route ties baseline at 76.0. We read this as evidence that BarExam needs a rewrite-vs-Snap-HyRE selector, not a single fixed route; the calibration controller picks the right family on its own slice but the policy that selects within the family is the open piece.

HousingQA (statutory entailment on top of jurisdiction scope). This is the cleanest controller exhibit. Baseline rag_state_filter sits at 60.5; HyRE-only is a negative control at 50.0 and fixed Snap-HyRE at 51.5; the verifier route reaches 74.5 on calibration and 76.0 held-out, a +14pp lift over the matched baseline. The state-filter casing fix (question-side California → corpus-side california) converted the filter from a parametric fallback to real retrieval: 0/200 empty-retrieval rows became 81/200 gold-retrieved at $k = 5$, a +8.0pp lift over generic top-5 (paired $p = 0.0440$). With jurisdiction scope handled by the metadata filter, the residual bottleneck is false-positive yes: HyRE biases the generator toward affirmative answers in cases where the controlling statute does not support entailment. The conservative yes/no verifier on top of state-filter retrieval clears that bottleneck without invoking a HyRE expansion, which is also why HousingQA's average call cost stays low.

CaseHOLD (answer-option conversion). The repaired rag_simple-vs-rag_snap_hyde_2call pair shows 69.5 → 72.0 on EM (+2.5pp, $b/c = 16/11$, $p = 0.4421$) while gold-passage retrieval rises from 16.0 to 47.0, a +31pp exposure gain. The signal is unmistakable: retrieval-side gains do not convert into

Benchmark	Active bottleneck	Diagnostic signal	Chosen route
BarExam	Query / legal-reasoning formulation	Rewrite vs. Snap-HyRE asymmetry; snap reasoning lifts above HyDE retrieval alone	Adaptive Snap-HyRE-v2 (rewrite vs. Snap-HyRE selector)
HousingQA	Jurisdiction scope + false-positive yes	Empty/near-zero state-match on baseline; verifier lifts over state-filter; fixed HyRE underperforms baseline	State filter + conservative yes/no verifier
CaseHOLD	Answer-option conversion	Gold-hit 16.0 → 47.0 but EM only +2.5pp (NS)	Diverse Snap-HyRE
SCALR	Method disagreement / candidate exposure	Top-1 collapse; top-5/top-10 tie; high HyRE-family disagreement	Disagreement arbitration / frontier replay

Table 5. Per-benchmark bottleneck reading, calibration evidence, and the controller's chosen route.

answer-side gains, which we type as answer-option conversion. The controller responds with diverse Snap-HyRE, which produces multiple holding-shaped pseudo-passages so that the option-anchoring step has heterogeneous evidence to disambiguate against; held-out diverse Snap-HyRE reaches 78.0 vs. a 68.0 baseline (+10pp). The negative design point on the same slice is direct option-table prompting at 70.0, below both rag_rewrite (76.0) and diverse Snap-HyRE (78.0). The bottleneck is which option to anchor to, not whether to retrieve more.

Dataset	Baseline	HyRE-only	Fixed Snap-HyRE	Best route
BarExam	80.0	82.0	84.5	86.0
HousingQA	60.5	50.0	51.5	74.5
CaseHOLD	73.0	71.5	72.0	73.5
SCALR	74.0	74.0†	76.0	77.5

Table 6. Expanded method ladder on the calibration slice. Both pure HyRE and fixed Snap-HyRE fall below the matched baseline on HousingQA and CaseHOLD; the controller's best route dominates every fixed policy on every benchmark. †SCALR HyRE-only is reported from a length-capped generation pass (148/200 rows complete with no parse failures).

LegalBench-SCALR (method disagreement / candidate exposure). The top-k ablation on Llama 70B collapses top-1 to 59.5% while top-5 and top-10 tie at 77.0%; gold-passage hit rate rises top-5 → top-10 from 54.0 to 63.0 but answer accuracy is flat. The retrieval candidate set is therefore saturated past k=5 and the residual problem is method disagreement over a small reranked set of holding candidates. Calibration: the disagreement-arbitration controller reaches 77.5 vs. rag_simple 74.0. Held-out: the exact selected route ties the matched baseline at 80.0, but the frontier replay component on the same slice reaches 84.0. The headroom is real and the policy that should pick which candidate-set arbiter to invoke per row is the open routing-policy question on SCALR.

5.3. Negative controls: when fixed Snap-HyRE and HyRE-only hurt

A central piece of the headline is what the controller does not do. Both pure-HyRE and fixed Snap-HyRE underperform the matched baseline on the two benchmarks where the active bottleneck is not query formulation (Table 6).

HousingQA HyRE-only is 10.5pp below the matched baseline and fixed Snap-HyRE is 9.0pp below; on CaseHOLD HyRE-only is 1.5pp below baseline and fixed Snap-HyRE is 1.0pp below. These are not retrieval failures in the usual sense: the HyRE family delivers higher gold-passage exposure on CaseHOLD (16.0 → 47.0) and parametrically-coherent passages on HousingQA, but they are intervention-mismatch failures. Adding a HyDE-style expansion does not help when the active bottleneck is jurisdiction scope or answer-option conversion, and on HousingQA it actively biases the verifier toward affirmative answers. The controller avoids these losses precisely by reading the calibration signature before committing to a HyRE-family route.

5.4. Cross-model sanity (Llama 3.3 70B held-out)

We re-ran the held-out slice on Groq Llama 3.3 70B to check whether the controller's bottleneck readings carry across model families (Table 7). The transfer is partial: two of four routes carry the expected direction (HousingQA verifier, SCALR frontier), one route flips sign (BarExam adaptive Snap-HyRE-v2), and one route did not produce a usable detail log on the available run (CaseHOLD diverse-Snap-HyRE). The honest read is that the bottleneck labels appear to be model-stable on HousingQA and SCALR, while the BarExam route choice itself is model-conditional.

We treat Table 7 as cross-model coverage, not as a main result: the slice is N=50 per benchmark,

Dataset	Baseline	Selected route	Read
BarExam	76.0	72.0	Slice-unstable
HousingQA	44.0	60.0	Verifier transfers (+16pp)
CaseHOLD	66.0	—	No usable detail log on this run
SCALR	82.0	88.0	Frontier transfers (+6pp)

Table 7. Cross-model sanity on Groq Llama 3.3 70B, rows 200–249 (N=50 per benchmark). Two of four routes transfer directionally (HousingQA verifier, SCALR frontier); BarExam adaptive Snap-HyRE-v2 underperforms the matched baseline on this slice; the CaseHOLD diverse-Snap-HyRE cell did not produce a usable detail log on the available run.

BarExam exhibits the same route-instability we already document at Gemma scale, and the CaseHOLD diverse-Snap-HyRE cell on Llama did not produce a usable detail log on our run. The defensible read is that the controller's HousingQA-verifier and SCALR-frontier diagnoses are model-stable while BarExam's rewrite-vs-Snap-HyRE selector remains the open piece.

5.5. Scope of evaluation

The four-benchmark calibration uses N=200 rows per benchmark; the held-out slice uses N=50. Larger-N evaluations on each benchmark are in progress and will be reported in a future revision. The bottleneck readings reported here should therefore be taken as diagnostic-grade evidence at fixed slice size rather than as final benchmark rankings.

6. Discussion, Limitations, and Conclusion

6.1. Discussion: routing as the unit of analysis

The dominant framing in retrieval-augmented question answering, "one method wins," collapses several distinct failure modes into a single accuracy delta and produces recipes that do not transfer across benchmarks. Our bottleneck-aware view inverts that framing: routing is the unit of analysis, and each benchmark is characterized by which intervention helps, which is neutral, and which actively hurts. Under this view, negative controls are not embarrassments to be hidden but signal. Fixed Snap-HyRE underperforming baseline on HousingQA and CaseHOLD is not a failure of Snap-HyRE. It is positive evidence about which bottleneck each benchmark exposes (metadata scope and statutory entailment on HousingQA, answer-option conversion on CaseHOLD), and therefore a guide to the intervention each benchmark actually needs.

Reading the calibration ladder this way also relocates the methodological novelty of the Snap-HyRE/HyRE family. The two-call architecture is a small efficiency variant of HyDE (Gao et al., 2023); what is new is the snap-conditioned passage prompt, which yields a controlling-rule embedding rather than a question paraphrase, and the placement of that embedding inside a routing portfolio rather than as a universal default. The same HyDE-family intervention is positive on BarExam (calibration Snap-HyRE 86.0), neutral on LegalBench-SCALR (76.0 vs. baseline 74.0), and negative on HousingQA (51.5 vs. 60.5) and CaseHOLD (72.0 vs. 73.0). The accuracy signature is a property of the bottleneck, not the method.

The diagnostic protocol behind the bottleneck-aware diagnostic controller is dataset- and model-agnostic by construction: any benchmark slice combined with any language model and any retrieval stack produces a calibration trace whose signals can be read off without retraining or annotated complexity labels. This stands in contrast to learned routers (Asai et al., 2024; Jeong et al., 2024), which require either dataset-specific fine-tuning of reflection or critique tokens or a supervised notion of question complexity. The protocol is therefore a candidate measurement layer rather than a competing recipe; we expect it to compose with learned components in future systems rather than replace them.

6.2. Limitations

Several limitations qualify the scope of these claims.

1. Slice sizes. The calibration tables use N=200 per benchmark and the held-out tables N=50. The four legal benchmarks have full sizes 1,195 / 6,853 / 3,600 / 571. Larger-N evaluations are in progress on each benchmark and the numbers in this draft should be read as diagnostic-grade evidence at fixed slice size, not final benchmark rankings.

2. Cross-model coverage. The Llama 3.3 70B sanity sweep is N=50 per benchmark and only partially transfers: the HousingQA verifier and SCALR frontier routes hold up directionally, BarExam Snap-HyRE-v2 underperforms baseline (72.0 vs. 76.0), and one CaseHOLD diverse-HyRE cell did not produce a usable detail log on our run. Multi-architecture generalization is therefore weakly, not strongly, supported.

3. Rule-based controller. The bottleneck-aware diagnostic controller is rule and evidence-summary based rather than learned. A learned router could plausibly exploit calibration-trace features that the current rule set cannot read off directly, and the gap between the rule-based controller and a learned upper bound is itself an interesting open question.

4. Held-out instability. The held-out slice is rows 200–249 (N=50 per dataset). BarExam (76.0 selected vs. 90.0 rewrite) and SCALR (80.0 selected vs. 84.0 frontier) are route-unstable on this slice. The routing policy is informative but not finished; a rewrite-vs-HyRE selector and a frontier-aware SCALR rule are the obvious next steps.

5. Reference baselines. We do not yet compare against full Asai et al. (2024) Self-RAG, Yan et al. (2024) CRAG, or Wang et al. (2025) Speculative RAG on legal benchmarks. Those comparisons require either retraining or per-question critique infrastructure and are deferred to future work.

6.3. Conclusion

Legal RAG fails through bottleneck-typed mechanisms rather than through a single missing capability. A cheap calibration trace, a fixed set of signals collected on a 200-row slice per benchmark, is enough to expose which mechanism is active. A rule-based bottleneck-aware diagnostic controller that routes among baseline retrieval, query rewrite, Snap-HyRE/HyRE, state-filter retrieval, option grounding, a conservative verifier, disagreement arbitration, and reject/escalate lifts macro accuracy by roughly 6pp across the four legal benchmarks at a sub-2 LLM-call budget. Snap-HyRE is one route among many: useful where the bottleneck is query or legal-reasoning formulation, ineffective or harmful where the bottleneck is metadata scope or answer-option conversion. The contribution of this paper is therefore not a new universal recipe but a discipline, name and measure the bottleneck before paying for the intervention, and the diagnostic adaptation framework controller that operationalizes that discipline.

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A. Extended Result Tables

This appendix lays out every result row that appears in any main-paper table, then expands the bottleneck taxonomy with worked examples and preserves a BarExam Tier-3 full-corpus reference result.

The bold rows above are the load-bearing cells. The two negative controls (HyRE-only on HousingQA, fixed Snap-HyRE on HousingQA) underperform the matched baseline; treating Snap-HyRE-family interventions as universal would lose these rows. The bolded held-out lifts (+14pp on HousingQA, +10pp on CaseHOLD) carry the macro headline. The bolded route-unstable BarExam held-out cell (76.0 for the selected route while rag_rewrite reaches 90.0 on the same rows) is the open routing-policy gap discussed in Section 6. The cross-model transfers (HousingQA verifier, SCALR frontier) are partial: the BarExam adaptive-Snap-HyRE route does not transfer, and we omit the Llama 70B CaseHOLD diverse-Snap-HyRE cell because that particular run did not produce a usable detail log.

B. Bottleneck Taxonomy with Worked Examples

Table 2 in Section 3.4 compresses the eight bottleneck labels onto a single page. The expanded form below pairs each label with the cheapest calibration signal that identifies it, the intervention the controller selects, an example trace pattern from the actual calibration logs, and a one-sentence interpretation. "Trace" figures are taken directly from the cite-clean calibration rows in Table 8 (Gemma 4 26B-A4B, N=200); the held-out and Llama numbers are reported as cross-checks where relevant.

1. Retrieval depth. Signal. A catastrophic top-1 vs. top-5 drop on the baseline retrieval row: the answer accuracy collapses when only the single best reranked passage is shown, which is the diagnostic that the gold rule is present at rank ≤ 5 but not at rank 1. Intervention. Increase k and/or use a smarter passage selector upstream of the synthesizer. Trace pattern. Reference multi-hop slices show top-1 vs. top-5 gaps in excess of 10pp on baseline retrieval; the four legal benchmarks we run are not depth-catastrophic, which is one half of why the legal-RAG bottleneck taxonomy needs more than depth ablations to be informative. Interpretation. The depth bottleneck is the cleanest classical RAG failure but is empirically rare on legal multiple-choice / yes-no benchmarks; presenting it inside the taxonomy is what allows the controller to rule it out as the dominant signal on the four legal datasets and

route reasoning elsewhere.

2. Candidate-set size then saturated. Signal. Top-1 retrieval collapses (very low recall@1) yet top-5/top-10 retrieval ties; gold retrieval rate rises but the answer accuracy is approximately flat. Intervention. Restrict to a small reranked candidate set; do not spend additional LLM calls on retrieval reformulation. Trace pattern. LegalBench-SCALR baseline: recall@1 $\approx$ 0.52 but recall@5 = 1.0 and recall@10 = 1.0 on the same rows; the gold rule is reliably in the top-5 pool but the model does not always pick it as the single decisive holding (baseline 74.0% at N=200). Interpretation. On SCALR, retrieval has already saturated the useful pool; the lift has to come from candidate disambiguation, not from looking harder for the rule. This is exactly the gap the frontier and disagreement routes target.

3. Query / legal-reasoning formulation. Signal. Rewrite-vs-Snap-HyRE asymmetry on the calibration slice; snap-only reasoning lifts more than HyDE retrieval alone. Intervention. rag_rewrite or adaptive_snap_hyre_v2; the controller selects between them per row. Trace pattern. BarExam calibration: baseline 80.0, snap-only 85.5, HyDE-only 82.0, fixed Snap-HyRE 84.5, adaptive Snap-HyRE-v2 86.0. Held-out: rewrite 90.0 vs. adaptive-v2 76.0 on the same N=50 slice: rewrite wins this slice while the slice-level-selected Snap-HyRE route ties baseline. Interpretation. BarExam is the benchmark on which formulation choice (rewrite vs. snap-conditioned hypothetical passage) matters more than retrieval depth. The route is which kind of pre-retrieval text to produce, not whether to retrieve.

4. Metadata / jurisdiction scope. Signal. Empty-retrieval rate is high on generic top-k retrieval; an explicit metadata filter beats generic retrieval; near-zero state-match rate without the filter. Intervention. rag_state_filter (lowercased Chroma predicate over the statute state field). Trace pattern. HousingQA generic rag_simple (pre-fix) was 0/200 empty retrieval and the filter was a parametric fallback. The casing-repaired rag_state_filter reaches 60.5% at N=200 with 81/200 gold retrieved at k=5; raw rag_hyde on the same slice falls to 50.0%, and fixed Snap-HyRE to 51.5% (both negative controls). Interpretation. HousingQA's first-order bottleneck is jurisdiction scope, not retrieval depth or query rewriting. Once the state filter is in place, additional retrieval interventions degrade the result.

5. Statutory entailment / false-positive yes. Signal. A conservative yes/no verifier route lifts over the state-filter baseline; fixed Snap-HyRE underperforms baseline. The diagnostic is: when retrieval already finds the right statute, the model still over-predicts yes on entailment, and a verifier prompt that demands explicit statutory support flips the false positives. Intervention. adaptive_snap_hyre_housing_verifier (state filter + conservative yes/no verifier on top). Trace pattern. HousingQA calibration: state-filter 60.5 $\rightarrow$ verifier 74.5 (+14.0pp). Held-out: state-filter 62.0 $\rightarrow$ verifier 76.0 (+14.0pp). Llama 70B cross-model: state-filter 44.0 $\rightarrow$ verifier 60.0 (+16.0pp, transfers directionally). Interpretation. The HousingQA gain is the single most transferable controller win and is what positions the verifier policy as the strongest source-gated route in the portfolio.

6. Answer-option conversion. Signal. Gold-retrieval gains do not convert to answer gains; the model has the rule in its context window but does not pick the right one of five candidate holdings. Intervention. Diverse Snap-HyRE passages or option-table prompting; both produce more option spread for the synthesizer to disambiguate against. The direct option-table route is a clean negative on this slice and is reported as a design point, not as a positive route. Trace pattern. CaseHOLD: gold retrieval rises from 16.0% $\rightarrow$ 47.0% between rag_simple and rag_snap_hyde_2call while exact-match accuracy rises only 69.5% $\rightarrow$ 72.0% (+2.5pp, b/c=16/11, p=0.4421; job 58283). At N=200 calibration the matched baseline is 73.0% and diverse Snap-HyRE reaches 73.5%; held-out diverse Snap-HyRE reaches 78.0% vs. rewrite 76.0% and direct option-table 70.0% (job 67744). Interpretation. CaseHOLD's first-order bottleneck is not retrieval but the conversion from a topical passage to a discrete holding choice. Retrieval interventions help less than option-spread interventions, and telling the model the candidate holdings without a diverse retrieval pool collapses the answer distribution.

7. Method disagreement / candidate exposure. Signal. High row-level disagreement across the HyRE-family rows (two routes give different predicted holdings on the same rows), and an oracle row-level ceiling (the best-of-N across HyRE rows) that is not realized by simple ensembles. Empirically the gold rule is in the top-5 pool for almost every row, so the lift is about picking among candidates, not about candidate discovery. Intervention. Disagreement arbitration (adaptive_snap_hyre_disagreement_majority_prior) or a frontier replay selector (adaptive_snap_hyre_frontier). Trace

pattern. LegalBench-SCALR calibration: rag_snap_hyde_2call 76.0, disagreement arbitrator 77.5 (+1.5pp). Held-out: frontier 84.0 vs. exact-replay route 80.0 (+4.0pp); Llama 70B cross-model: frontier 88.0 vs. baseline 82.0 (+6.0pp, transfers directionally). Interpretation. The lift exists but the controller currently picks the wrong route on the held-out SCALR slice. This is reported as an open routing-policy gap, not as a closed result.

8. Parametric floor / depth-flat. Signal. Top-1 $\approx$ top-5 $\approx$ llm_only; the snap-only row matches RAG; the gold-conditional accuracy on the few rows where gold is retrieved is not appreciably higher than on rows where gold is missing. Intervention. Trust priors. Use snap reasoning to anchor the final answer, not to drive retrieval. Trace pattern. BarExam: Acc | gold-missing = 80.5% vs. Acc | gold-retrieved = 60.0% on baseline retrieval (the gold-conditional row inverts on BarExam, which is a signature of the parametric floor); BarExam llm-only is historically $\approx$ 79.8% on this model (see Appendix C); the snap-only and snap-anchored routes are within $\pm$2pp of each other. Interpretation. BarExam reasoning is co-located with the parametric floor: useful interventions are about anchoring the model's existing knowledge with a snap commit, not about feeding it more retrieved text. The depth-flat label and the formulation label both apply to BarExam; the controller treats formulation as the dominant slice-level signal because rewrite-vs-Snap-HyRE asymmetry is the largest cheap discriminator.

C. BarExam Tier 3 Full-Corpus Reference Result

This section preserves a pre-pivot full-corpus BarExam reference result. The diagnostic-adaptation contribution of this paper does not depend on it; the result is reported here because it is the closest thing we have to a full-corpus number on any legal benchmark, and because it supports the BarExam-route bottleneck story (query / legal-reasoning formulation) at scale.

This is the strongest single-benchmark statistical result in the project, and it is the cross-size robustness check for the BarExam route the controller selects in the main paper. On the BarExam Tier 3 slice (N=1,195, the full BarExam test set), rag_snap_hyde reaches 81.17% vs. rag_simple 78.08% on Gemma 4 26B-A4B (a +3.09pp paired lift, b/c=124/87, McNemar p=0.0130); the same comparison on Gemma 4 E4B is rag_snap_hyde 62.18% vs. rag_simple 58.49% (+3.69pp, b/c=172/128, p=0.0129). Both deltas are positive, statistically significant, and reproduce at two model sizes on the same benchmark. The full-corpus positive on BarExam is consistent with the calibration ladder: the controller routes BarExam to adaptive Snap-HyRE-v2 precisely because calibration shows the same lift in miniature (86.0 vs. 80.0 at N=200).

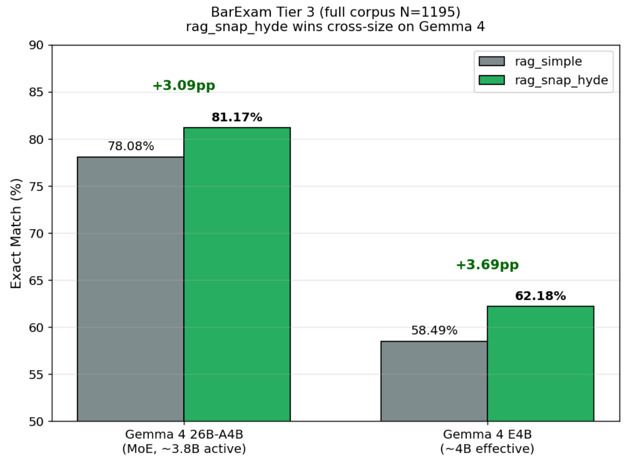

Figure 6. BarExam Tier 3 full-corpus cross-size lift. rag_snap_hyde beats rag_simple at both Gemma 4 26B-A4B (+3.09pp, p=0.0130) and Gemma 4 E4B (+3.69pp, p=0.0129) on the full N=1,195 slice, which is the cross-size robustness check for the BarExam route the controller selects in the main paper.

The full-corpus result also makes a point that the calibration and held-out tables cannot make on their own: the BarExam route choice is not a calibration-fitting artifact. The same intervention that wins the N=200 slice also wins the N=1,195 slice, on the same model. The limit of this evidence is that it is single-benchmark: the four-benchmark calibration shows that fixed Snap-HyRE underperforms baseline on HousingQA and CaseHOLD (negative controls in Table 8), so the controller's value is precisely that it withholds Snap-HyRE where the negative controls predict it will hurt and applies it where this full-corpus result predicts it will help.

The BarExam single-passage golden_passage control is the row that most clearly motivates the depth-flat / parametric-floor reading. Providing the labeled gold passage reaches only 78.66%, below both llm_only (79.75%) and snap_only_in_final (80.59%). Mechanism: a labeled passage on a multi-issue bar-exam fact pattern can be topically correct but not decisive for the question asked, so a model that anchors on it can be pulled away from the right rule. Snap-first reasoning gives the model an initial legal frame before evidence reconciliation, which is why snap_only_in_final beats the golden-passage con-

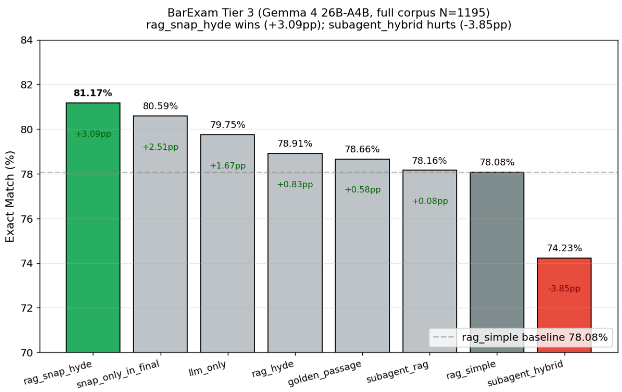

Figure 7. BarExam Tier 3 full-corpus method matrix for Gemma 4 26B-A4B (N=1,195). Numerical entries are reproduced in Table 9; visualizing the matrix makes clear that rag_snap_hyde, snap_only_in_final, and llm_only cluster within ≈ 1.5pp of each other, diagnostic of the depth-flat / parametric-floor co-bottleneck on this benchmark.

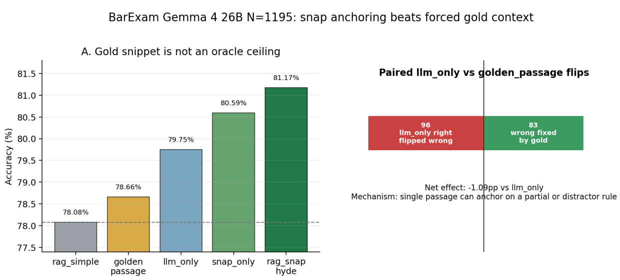

Figure 8. BarExam single-passage golden_passage control is a noisy diagnostic, not an oracle ceiling. The labeled passage reaches only 78.66%, below llm_only (79.75%) and snap_only_in_final (80.59%). The mechanism story is in the text below.

trol on this benchmark. The general lesson the appendix carries forward into the main paper is that "oracle"-style controls on legal multiple-choice are diagnostic of how the model uses a single passage, not of a retrieval ceiling.

Table 8. Complete result table used in the main paper. Calibration rows are Gemma 4 26B-A4B on OpenRouter over rows 0–199 (N=200); held-out rows are the same model over rows 200–249 (N=50); cross-model rows are Groq Llama 3.3 70B over rows 200–249. Notable cells are annotated.

Method	Dataset	Slice	N	Acc. (%)	Calls	Note
Calibration (Gemma 4 26B-A4B, rows 0–199)						
rag_simple	BarExam	calib	200	80.0	1.00	matched baseline
rag_state_filter	HousingQA	calib	200	60.5	1.00	matched baseline (ju
rag_simple	CaseHOLD	calib	200	73.0	1.00	matched baseline
rag_simple	SCALR	calib	200	74.0	1.00	matched baseline
snap_only_in_final	BarExam	calib	200	85.5	2.00	parametric snap; no
snap_only_in_final	HousingQA	calib	200	55.0	2.00	parametric snap; no
snap_only_in_final	CaseHOLD	calib	200	72.5	2.00	parametric snap; no
snap_only_in_final	SCALR	calib	200	72.5	2.00	parametric snap; no
rag_rewrite	BarExam	calib	200	82.0	2.00	legal query rewrite
rag_rewrite	HousingQA	calib	50	58.0	2.00	N=50 control
rag_rewrite	CaseHOLD	calib	50	72.0	2.00	N=50 control
rag_rewrite	SCALR	calib	50	76.0	2.00	N=50 control
rag_hyde	BarExam	calib	200	82.0	2.00	HyRE-only retrieval
rag_hyde	HousingQA	calib	200	50.0	2.00	negative control
rag_hyde	CaseHOLD	calib	200	71.5	2.00	weak negative
rag_hyde	SCALR	calib	200	74.0	2.00	length-capped pass
rag_snap_hyde_2call	BarExam	calib	200	84.5	2.00	fixed Snap-HyRE
rag_snap_hyde_2call	HousingQA	calib	200	51.5	2.00	negative control
rag_snap_hyde_2call	CaseHOLD	calib	200	72.0	2.00	weak negative
rag_snap_hyde_2call	SCALR	calib	200	76.0	2.00	fixed Snap-HyRE
adaptive_snap_hyre_v2	BarExam	calib	200	86.0	2.00	best calibration route
adaptive_snap_hyre_housing_verifier	HousingQA	calib	200	74.5	1.00	best calibration route
adaptive_snap_hyre_diverse	CaseHOLD	calib	200	73.5	2.00	best calibration route
adaptive_snap_hyre_disagreement_majority_prior	SCALR	calib	200	77.5	0.19	best calibration route
Held-out (Gemma 4 26B-A4B, rows 200–249)						
rag_simple	BarExam	held	50	76.0	1.00	held-out baseline
rag_state_filter	HousingQA	held	50	62.0	1.00	held-out baseline
rag_simple	CaseHOLD	held	50	68.0	1.00	held-out baseline
rag_simple	SCALR	held	50	80.0	1.00	held-out baseline
rag_rewrite	BarExam	held	50	90.0	2.00	rewrite dominates Ba
rag_rewrite	HousingQA	held	50	58.0	2.00	rewrite below baselin
rag_rewrite	CaseHOLD	held	50	76.0	2.00	strong rewrite contro
rag_rewrite	SCALR	held	50	78.0	2.00	rewrite below baselin
adaptive_snap_hyre_v2	BarExam	held	50	76.0	2.00	ties baseline; route-u
adaptive_snap_hyre_housing_verifier	HousingQA	held	50	76.0	2.00	+14pp over baseline
adaptive_snap_hyre_diverse	CaseHOLD	held	50	78.0	2.00	+10pp over baseline
adaptive_snap_hyre_frontier	SCALR	held	50	84.0	2.00	frontier component r
adaptive_snap_hyre_disagreement_majority_prior	SCALR	held	50	80.0	0.19	exact-replay route tie
adaptive_snap_hyre_option_table	CaseHOLD	held	50	70.0	2.00	negative design point
Cross-model held-out (Groq Llama 3.3 70B, rows 200–249)						
rag_simple	BarExam	Llama held	50	76.0	1.00	cross-model baseline
adaptive_snap_hyre_v2	BarExam	Llama held	50	72.0	2.00	does not transfer
rag_state_filter	HousingQA	Llama held	50	44.0	1.00	cross-model baseline
adaptive_snap_hyre_housing_verifier	HousingQA	Llama held	50	60.0	2.00	transfers (+16pp)
rag_simple	CaseHOLD	Llama held	50	66.0	1.00	cross-model baseline
rag_simple	SCALR	Llama held	50	82.0	1.00	cross-model baseline
adaptive_snap_hyre_frontier	SCALR	Llama held	50	88.0	2.00	transfers (+6pp)
Dataset-specific gating references						
rag_state_filter	HousingQA	calib (k=5)	200	61.5	1.00	casing-aware state fil
rag_state_filter	HousingQA	calib (k=10)	200	62.5	1.00	deeper retrieval, +1p
rag_simple	CaseHOLD	paired pair	200	69.5	1.00	paired with two-call
rag_snap_hyde_2call	CaseHOLD	paired pair	200	72.0	2.00	gold-hit 16%→47%;

Table 9. BarExam Gemma 4 26B-A4B Tier 3 method matrix (N=1,195). Reproduced from the post-fix compiled results table.

Method	Acc. (%)	Δ (pp)	Note
rag_snap_hyde	81.17	+3.09	winner
snap_only_in_final	80.59	+2.51	
llm_only	79.75	+1.67	
rag_hyde	78.91	+0.83	
golden_passage	78.66	+0.58	noisy oracle
subagent_rag	78.16	+0.08	
rag_simple	78.08	—	baseline
subagent_hybrid	74.23	−3.85	negative